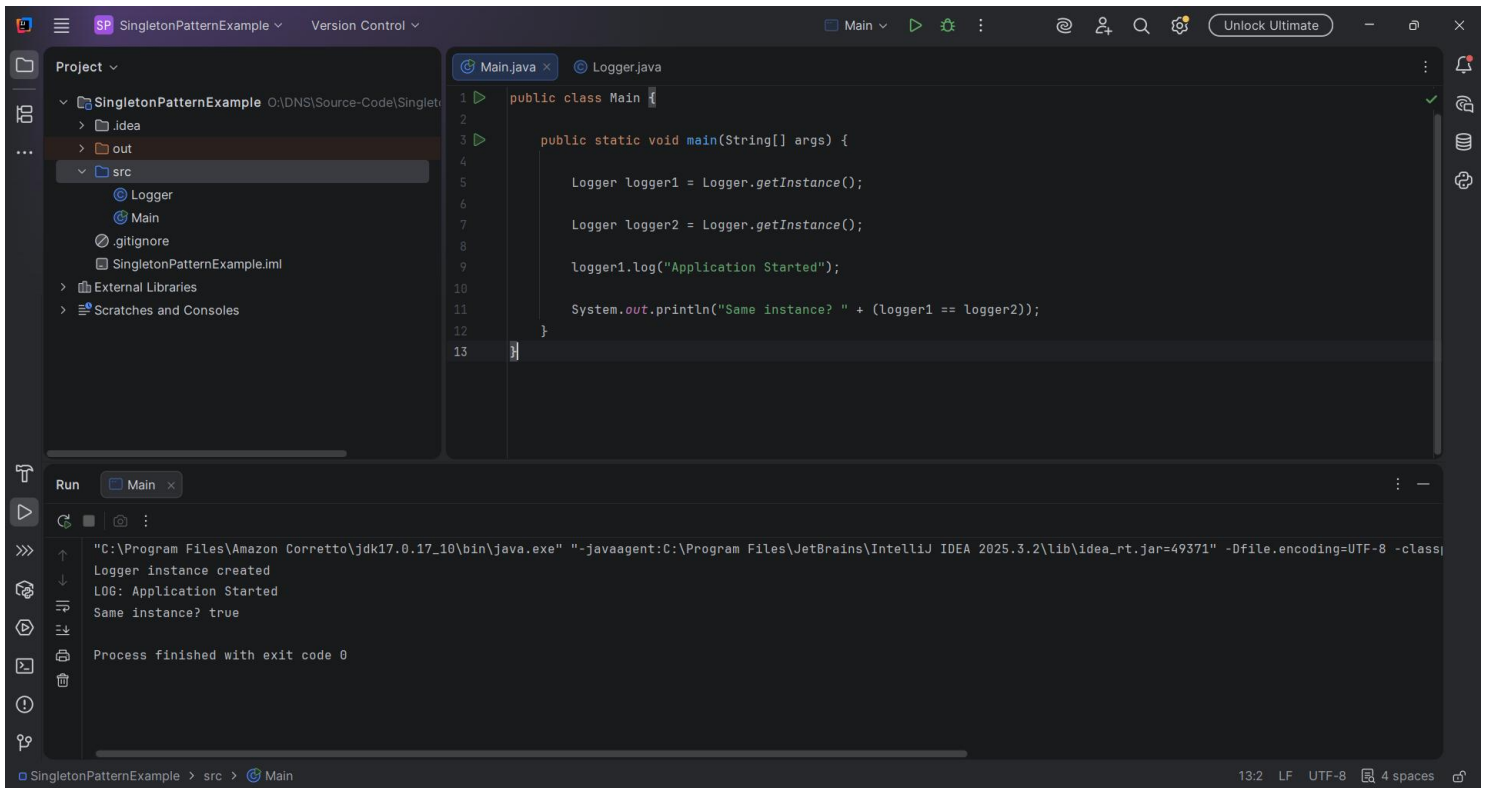


Solutions :-

Exercise-1

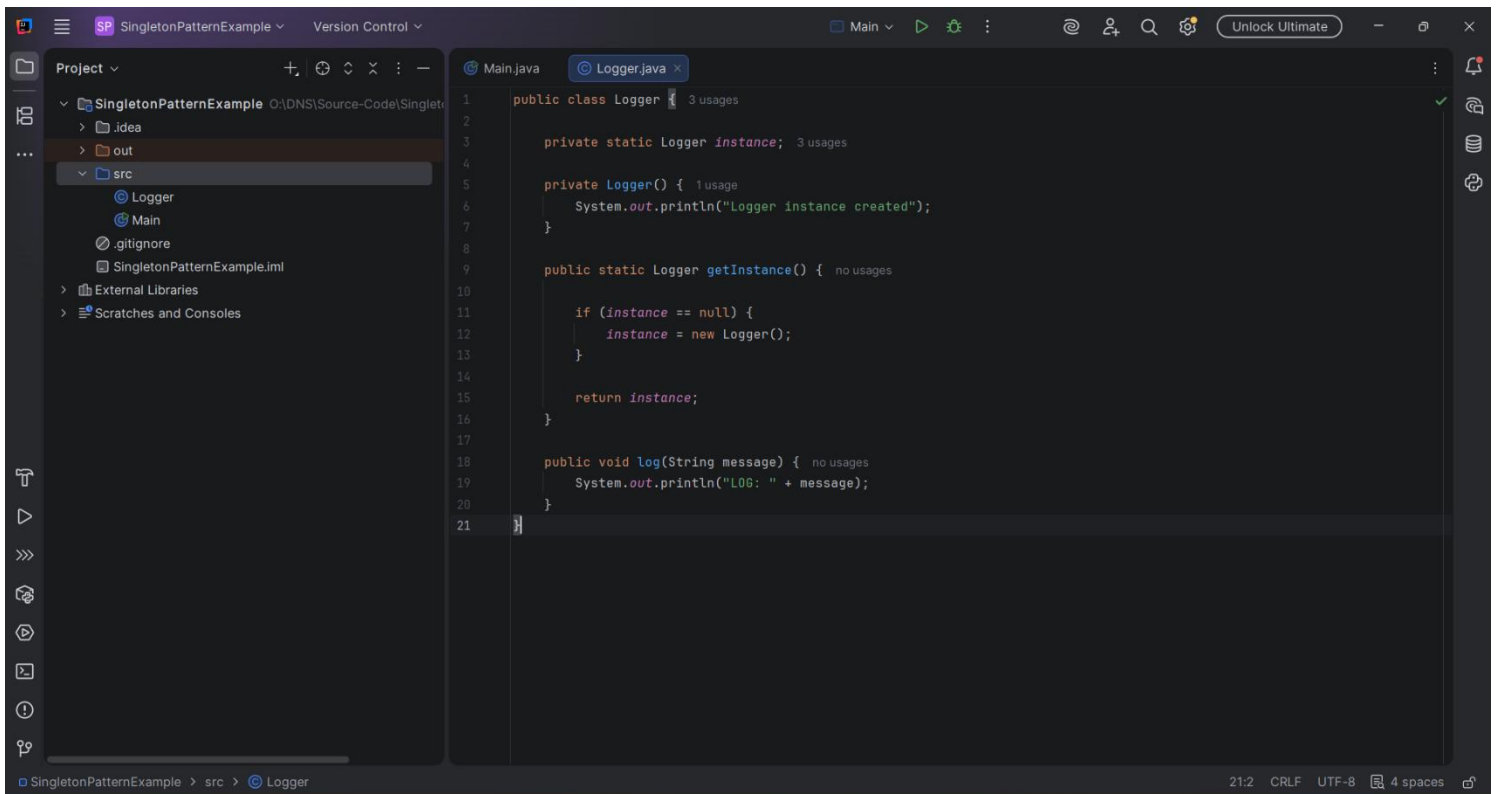


```
1 public class Main {
2
3     public static void main(String[] args) {
4
5         Logger logger1 = Logger.getInstance();
6
7         Logger logger2 = Logger.getInstance();
8
9         logger1.log("Application Started");
10
11         System.out.println("Same instance? " + (logger1 == logger2));
12     }
13 }
```

Run

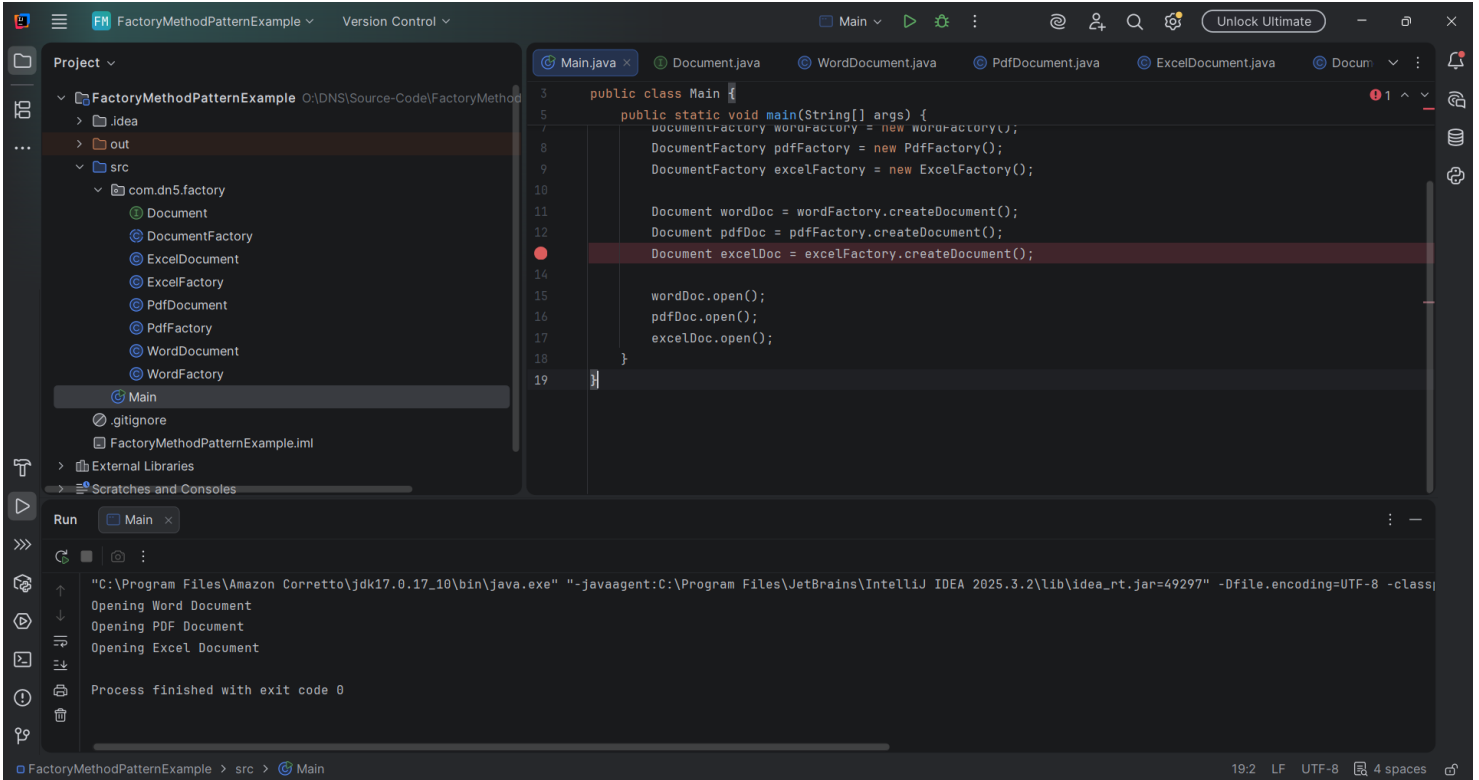
```
"C:\Program Files\Amazon Corretto\jdk17.0.17_10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=49371" -Dfile.encoding=UTF-8 -classpath
Logger instance created
LOG: Application Started
Same instance? true

Process finished with exit code 0
```

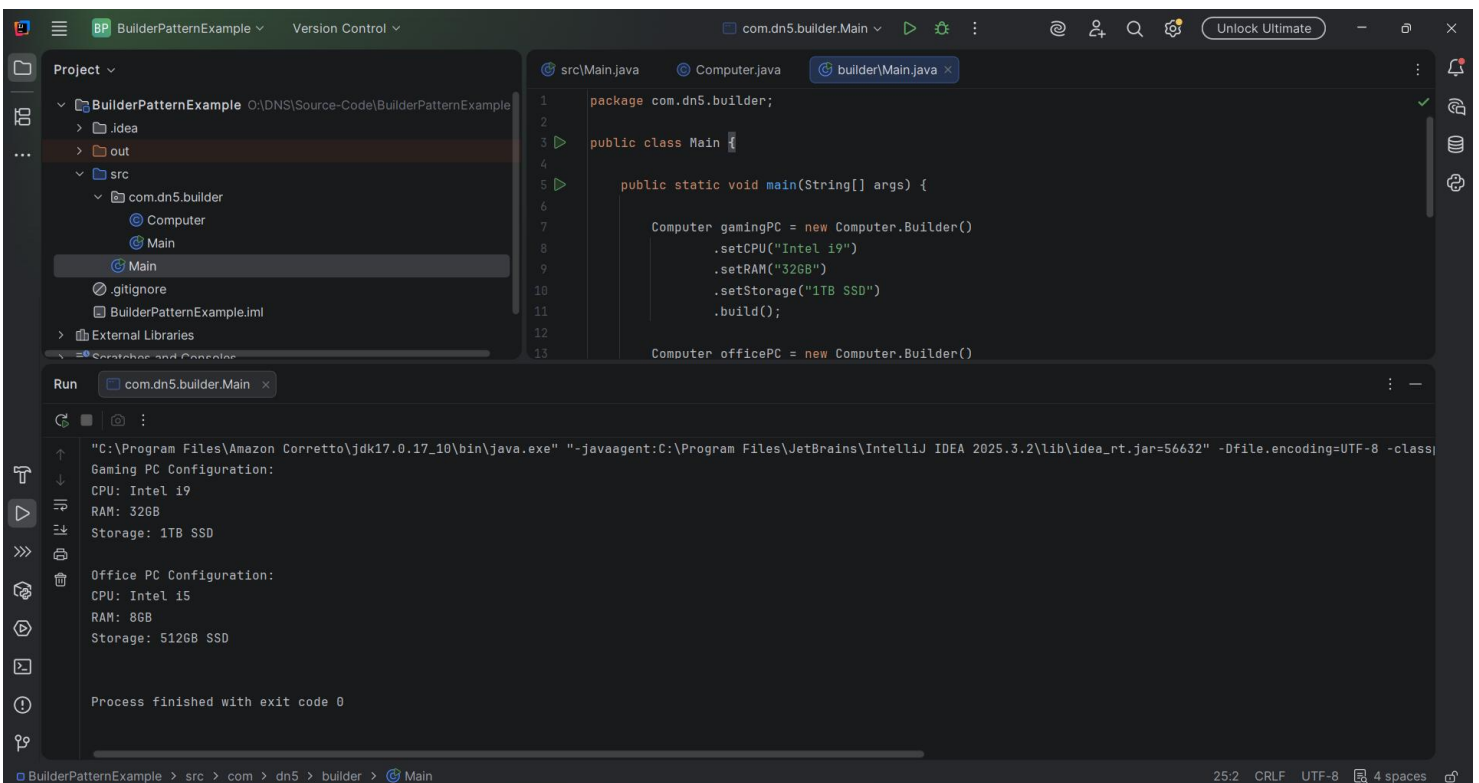


```
1 public class Logger {
2
3     private static Logger instance;
4
5     private Logger() {
6         System.out.println("Logger instance created");
7     }
8
9     public static Logger getInstance() {
10
11         if (instance == null) {
12             instance = new Logger();
13         }
14
15         return instance;
16     }
17
18     public void log(String message) {
19         System.out.println("LOG: " + message);
20     }
21 }
```

Exercise-2



Exercise-3



Exercise-4

```
package com.dn5.adapter;

public class Main {

    public static void main(String[] args) {

        PaymentProcessor paypal =
            new PayPalAdapter(new PayPalGateway());

        PaymentProcessor stripe =
            new StripeAdapter(new StripeGateway());

        paypal.processPayment( amount: 1000);

        stripe.processPayment( amount: 2500);

    }

}
```

Run Main

```
"C:\Program Files\Amazon Corretto\jdk17.0.17_10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55952" -Dfile.encoding=UTF-8 -classpath
PayPal Payment of Rs.1000.0 processed.
Stripe Payment of Rs.2500.0 processed.

Process finished with exit code 0
```

Exercise-5

```
package com.dn5.decorator;

public class Main {

    public static void main(String[] args) {

        Notifier notifier =
            new SlackNotifierDecorator(
                new SMSNotifierDecorator(
                    new EmailNotifier()
                )
            );

        notifier.send("Payment Successful");

    }

}
```

Run Main

```
"C:\Program Files\Amazon Corretto\jdk17.0.17_10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=50063" -Dfile.encoding=UTF-8 -classpath
Email Notification: Payment Successful
SMS Notification: Payment Successful
Slack Notification: Payment Successful

Process finished with exit code 0
```

Exercise-6

```
3 public class Main {
5     public static void main(String[] args) {
9         System.out.println("Image object created");
10
11         System.out.println("\nFirst Display:");
12
13         image.display();
14
15         System.out.println("\nSecond Display:");
16
17         image.display();
18     }
19 }
```

Run Main

```
"C:\Program Files\Amazon Corretto\jdk17.0.17_10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=50068" -Dfile.encoding=UTF-8 -classpath
Image object created

First Display:
Loading image from remote server: student_photo.jpg
Displaying image: student_photo.jpg

Second Display:
Displaying image: student_photo.jpg

Process finished with exit code 0
```

Exercise-7

```
3 public class Main {
5     public static void main(String[] args) {
11
12         Observer webUser =
13             new WebApp( user: "Investor");
14
15         stockMarket.registerObserver(mobileUser);
16
17         stockMarket.registerObserver(webUser);
18
19         System.out.println("Stock Price Changed\n");
20
21         stockMarket.setStockPrice( stockName: "TCS", price: 4200);
22     }
23 }
```

Run Main

```
"C:\Program Files\Amazon Corretto\jdk17.0.17_10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=64147" -Dfile.encoding=UTF-8 -classpath
Stock Price Changed

Mobile App [Sahil] : TCS updated to Rs.4200.0
Web App [Investor] : TCS updated to Rs.4200.0

Process finished with exit code 0
```

Exercise-8

```
3 public class Main {
5     public static void main(String[] args) {
11         new CreditCardPayment();
12     }
13
14     paymentContext.executePayment( amount: 5000);
15
16     paymentContext =
17         new PaymentContext(
18             new PayPalPayment()
19         );
20
21     paymentContext.executePayment( amount: 2500);
22 }
23 }
```

Run Main x

```
"C:\Program Files\Amazon Corretto\jdk17.0.17_10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55472" -Dfile.encoding=UTF-8 -class
Paid Rs.5000.0 using Credit Card
Paid Rs.2500.0 using PayPal

Process finished with exit code 0
```

StrategyPatternExample > src > com > dn5 > strategy > Main

Exercise-9

```
3 public class Main {
5     public static void main(String[] args) {
12         Command lightOn =
13             new LightOnCommand(light);
14
15         RemoteControl remote =
16             new RemoteControl();
17
18         remote.setCommand(lightOn);
19         remote.pressButton();
20
21         remote.setCommand(lightOff);
22         remote.pressButton();
23     }
24 }
```

Run Main x

```
"C:\Program Files\Amazon Corretto\jdk17.0.17_10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=54858" -Dfile.encoding=UTF-8 -class
Light is ON
Light is OFF

Process finished with exit code 0
```

CommandPatternExample > src > com > dn5 > command > Main

Exercise-10

